stg — EKS cluster svp-stg

ap-southeast-1 · 2 AZs, each with a public (NAT gateway only), an edge (the internal ALB only), a private and a database subnet · one NAT gateway, in the first AZ, for every private subnet · system node group on-demand and Karpenter's nodes (Spot), one icon each per AZ · API endpoint private only, reached through Systems Manager port forwarding to the SSM bastion · EBS volumes in the AZ of their pod's node · VPC flow logs, WAF logs and CloudFront access logs are written to S3 buckets of the svp-log-archive account

Database subnets, no route to the Internet: RDS single-AZ · Amazon MSK one broker per AZ, SASL/SCRAM-SHA-512 over TLS · ElastiCache for Valkey one node · AWS picks the AZs of RDS and ElastiCache (drawn from the first) · CloudWatch alarms of RDS, MSK and ElastiCache notify an SNS topic encrypted with a customer managed key, which emails the learner · KMS, Secrets Manager, CloudWatch, SNS, SQS, EventBridge regional

Kubernetes: a frame per namespace, an icon per workload with its replicas · the scheduler puts each pod on any node of any AZ, so the row of a frame or an icon says nothing about where its pods run (svp is drawn across the first two AZs to show it) · third-party tools as text · EKS add-ons (CoreDNS, kube-proxy, VPC CNI) not drawn; the GuardDuty runtime agent add-on reports through the guardduty-data interface endpoint

prd — EKS cluster svp-prd

ap-southeast-1 · 3 AZs, each with a public (NAT gateway only), an edge (the internal ALB only), a private and a database subnet · one NAT gateway per AZ · system node group on-demand and Karpenter's nodes (on-demand), one icon each per AZ · API endpoint private only, reached through Systems Manager port forwarding to the SSM bastion · EBS volumes in the AZ of their pod's node · VPC flow logs, WAF logs and CloudFront access logs are written to S3 buckets of the svp-log-archive account

Database subnets, no route to the Internet: RDS Multi-AZ primary/standby · Amazon MSK one broker per AZ, SASL/SCRAM-SHA-512 over TLS · ElastiCache for Valkey primary and 1 replica(s), Multi-AZ, automatic failover · AWS picks the AZs of RDS and ElastiCache (drawn from the first) · CloudWatch alarms of RDS, MSK and ElastiCache notify an SNS topic encrypted with a customer managed key, which emails the learner · KMS, Secrets Manager, CloudWatch, SNS, SQS, EventBridge regional

Kubernetes: a frame per namespace, an icon per workload with its replicas · the scheduler puts each pod on any node of any AZ, so the row of a frame or an icon says nothing about where its pods run (svp is drawn across the first two AZs to show it) · third-party tools as text · EKS add-ons (CoreDNS, kube-proxy, VPC CNI) not drawn; the GuardDuty runtime agent add-on reports through the guardduty-data interface endpoint

AWS Cloud

AWS Account svp-prd

Region ap-southeast-1

Web and API: viewer → CloudFront (web ACL, ACM certificate in us-east-1) → CloudFront VPC origin → internal ALB in the edge subnets, which only CloudFront may reach (security group on the CloudFront origin-facing prefix list); the ALB targets Envoy's pods directly (target group svp-prd-envoy, filled by the TargetGroupBinding), Envoy routes by Host to web and gateway · Media: CloudFront → S3 media through origin access control, pay-as-you-go · Route 53 aliases to the three distributions · access logs of the three distributions and logs of the three web ACLs go to the log archive account

EKS control plane · AWS managed

VPC 10.0.0.0/16

DNS

Route 53 · aws.<domain>
alias A/AAAA, CAA
DNSSEC, query log

Internet gateway

WAF web ACL
svp-prd-web

CloudFront svp-prd-web
aws.<domain>

Users

WAF web ACL
svp-prd-api

CloudFront svp-prd-api
api.aws.<domain>

GET HLS

AZ ap-southeast-1a

Public subnet 10.0.224.0/24

NAT gateway

Edge subnet 10.0.240.0/23

ALB svp-prd-edge
internal, HTTPS 443

VPC origin

Private subnet 10.0.0.0/19

EBS gp3 · PVCs

SSM bastion · t4g.nano

443 · API

Interface endpoints
16 AWS APIs, ENIs

443

System node group
on-demand

Karpenter node
on-demand

EKS cluster svp-prd

envoy-gateway-system

Service svp-envoy
ClusterIP

Envoy proxy
TargetGroupBinding

HTTPS 10443

cert-manager

internal CA svp-ca
(Envoy's certificate)

argocd · argo-rollouts · kyverno · keda · external-secrets

Argo CD, Argo Rollouts, Kyverno, KEDA, External Secrets

svp

web
x2 · Rollout

gateway
x2–4 · Rollout, HPA

user-service
x2

video-service
x2

engagement-service
x2

feed-service
x2

media-worker
x1–2 · KEDA

Database subnet 10.0.192.0/22

9096 · SASL/SCRAM-SHA-512, TLS

MSK broker

6379 · TLS

ElastiCache Valkey primary

5432 · TLS

RDS PostgreSQL primary

AZ ap-southeast-1b

Public subnet 10.0.225.0/24

NAT gateway

Edge subnet 10.0.242.0/23

registers targets

Private subnet 10.0.32.0/19

EBS gp3 · PVCs

volumes

System node group
on-demand

Karpenter node
on-demand

Interface endpoints
16 AWS APIs, ENIs

kube-system

AWS Load Balancer Controller,
EBS CSI driver, Karpenter

svp-data

svp-ops x1

kafka-init · Job

pg-bootstrap · Job

svp-obs

- Prometheus agent x2 → AMP
- OTel Collector
- ACK prometheusservice
- Grafana Operator
- Pyroscope, GlitchTip

Database subnet 10.0.196.0/22

MSK broker

ElastiCache Valkey replica

RDS standby

AZ ap-southeast-1c

Public subnet 10.0.226.0/24

NAT gateway

Edge subnet 10.0.244.0/23

EBS gp3 · PVCs

Private subnet 10.0.64.0/19

System node group
on-demand

Karpenter node
on-demand

Interface endpoints
16 AWS APIs, ENIs

Database subnet 10.0.200.0/22

MSK broker

S3 gateway endpoint

runtime events

GuardDuty interface endpoint
guardduty-data, 443 from the VPC

WAF web ACL
svp-prd-media

CloudFront svp-prd-media
media.aws.<domain>

OAC

Regional services

S3 media

S3 uploads
GuardDuty malware scan

PutObject media/*

GetObject

Role svp-prd-mediaconvert
GetObject uploads/*, PutObject media/*

assumes

Secrets Manager
RDS master, svp/prd/*,
AmazonMSK_*, ecr-pullthroughcache/*

KMS keys, customer managed
Terraform state, RDS, MSK,
SNS of the alarms, ECR,
observability
EKS Secrets

CloudWatch: alarms, Logs
control plane, RDS
MSK brokers, Valkey slow log

alarm, OK

SNS svp-prd-alarms
email to the learner

EventBridge rules
Spot interruption, rebalance,
instance state

SQS svp-prd-karpenter
read by Karpenter

Systems Manager
port forwarding to the bastion

ACM · ALB certificate, ap-southeast-1
viewer certificates in us-east-1

CreateJob, GetJob, Probe

ReceiveMessage

MediaConvert queue svp-prd
on-demand

job state change

EventBridge rule
svp-prd-mediaconvert-job-state

COMPLETE, ERROR

SQS svp-prd-transcode-events
read by media-worker

redrive

SQS svp-prd-
transcode-events-dlq

CloudWatch alarm
svp-prd-transcode-events-dlq

failed delivery

- media-worker runs its mediaconvert command here (same image and release as the ffmpeg worker): Kafka video.uploaded in, MediaConvert jobs out, transcode.completed or transcode.failed back
- MediaConvert reads uploads and writes media as the role; its job-state events (COMPLETE, ERROR) reach media-worker through EventBridge and SQS; an event EventBridge cannot deliver and a message media-worker cannot handle land in the DLQ, which pages through SNS
- media-worker calls MediaConvert and SQS through the interface endpoints and S3 through the gateway endpoint; no NAT

ECR svp-prd
svp-backend/*, svp-frontend/web
cache: ghcr, docker-hub, quay, k8s, ecr-public

scan

Inspector
enhanced scan of the cache repositories

AMP workspace svp-prd
metrics, rules, alerts

PromQL

AMG svp-prd
IAM Identity Center sign-in

Logs Insights

CloudWatch Logs
/svp/prd/otel, aws/spans

spans

X-Ray
Transaction Search

- ECR, AMP, AMG, the application log group and the X-Ray settings are in the root of the account, so they outlive the session; keys alias/svp-prd-ecr and alias/svp-prd-observability
- Collector and Prometheus agent write to AMP, CloudWatch Logs and X-Ray; nodes pull images from ECR, third-party images through its pull-through cache
- Interface endpoints: private DNS, 443 from the VPC, ENIs in the private subnets of the first 3 AZ(s); GitHub, Sigstore, Telegram, upstream registries, the Pricing API and AMG still go through NAT

KMS alias/svp-prd-uploads
SSE-KMS and Bucket Key
of S3 uploads

KMS alias/svp-prd-backup
key of the vault svp-prd

AWS Backup vault svp-prd
no lock

backup, restore jobs

Role svp-prd-backup
managed policies of AWS Backup

CloudWatch alarm
svp-prd-backup-job-failed

CloudWatch alarm
svp-prd-backup-copy-failed

copy job

- Backup policy svp-backup-prd of the organization puts the plan svp-prd here: rule daily, cron(0 18 * * ? *) UTC, the RDS instance and the media and uploads buckets tagged Backup=daily are its selection, run as the role
- Recovery points live 35 days here (emptied at the end of each session) and 35 days in the copy
- Both alarms notify SNS svp-prd-alarms (ticket)

POST presigned form

flow logs

logs

AWS Account svp-log-archive · buckets written by this env

S3 aws-waf-logs-svp
logs of the three web ACLs, us-east-1

S3 svp-cloudfront
CloudFront access logs, us-east-1

SSE-KMS with alias/svp-log-archive · blocked from public access · written by service principals of the organization (aws:SourceOrgID)

S3 svp-flowlogs
VPC flow logs

AWS Account svp-backup · copy of the prd backups

Only the management account writes to svp-backup (the learner has read-only SSO) · the vault access policy lets svp-prd copy into the vault and nothing else, so an administrator of the env cannot delete the copies

KMS alias/svp-backup-copy
key of both copy vaults

AWS Backup vault svp-prd-copy
governance lock 7–35 days

org — organization, accounts and security flows

The management account holds the organization, IAM Identity Center and the organization trail · accounts in OUs as in the AWS Security Reference Architecture: Security (log archive, security tooling, backup) and Workloads (Prod, NonProd) · every account also runs the baseline (S3 Block Public Access, EBS default encryption, Config recorder to svp-config unless config_recording is off in aws/paid-features.json)

Green: logs to the log archive account and backup copies to the backup account · red: detection and events to the security account, then SNS and email · policies on the root reach every member account (SCP, RCP) or every account (declarative, tag)

AWS Organization · root

On the root: service control policies
- SCP **svp-deny-leave-organization**
- SCP **svp-protect-cloudtrail**
- SCP **svp-deny-other-regions**
- SCP **svp-protect-baseline**
- SCP **svp-deny-iam-users**

On the root: resource control, declarative, tag
- RCP **svp-data-perimeter**
- Declarative policy **svp-ec2-baseline**
- Tag policy **svp-tags**

Organization features
- Centralized root access: member accounts hold no root credentials
- Delegated administrator svp-security: GuardDuty, Security Hub, Config, IAM Access Analyzer, Inspector (ECR scanning of every member account with inspector_ecr always)
- Baseline roots apply from the management account through OrganizationAccountAccessRole
- AWS Backup: cross-account backup and the RDS, S3 opt-in set here; backup policies by OU, no delegated administrator

Management account

AWS Organizations
root, OUs, member accounts

IAM Identity Center
SvpAdmin, SvpReadOnly, SvpLogReader
platform groups svp-platform-admins, -viewers

CloudTrail organization trail
management events, S3 data events of prd

EventBridge svp-security-* rules
forwarder role

OU Security

svp-log-archive

S3 svp-cloudtrail
organization trail, Object Lock

S3 svp-config
recorder of every account with config_recording always

KMS alias/svp-log-archive
a key in each bucket region

S3 svp-flowlogs
VPC flow logs

S3 aws-waf-logs-svp
WAF logs, us-east-1

S3 svp-guardduty
finding export

S3 svp-cloudfront
CloudFront access logs, us-east-1

svp-security · delegated administrator

Config aggregator
whole organization

GuardDuty
organization, 2 regions

findings

Security Hub
FSBP, CIS v5

IAM Access Analyzer
organization

Amazon Inspector
delegated admin, ECR only

EventBridge bus
svp-security-events

SNS svp-security-alarms

email

Learner
email

svp-backup

Copies of the recovery points of stg and prd, in vaults with a governance lock, encrypted with their own key · each vault accepts copies from its env account only

KMS alias/svp-backup-copy
key of both copy vaults

AWS Backup vault svp-stg-copy
governance lock 1–7 days

AWS Backup vault svp-prd-copy
governance lock 7–35 days

copy job

copy job

OU Workloads

OU Prod

svp-prd

VPC flow logs

AMG svp-prd
ADMIN svp-platform-admins
VIEWER svp-platform-viewers

WAF web ACL logging
aws-waf-logs-svp

AWS Backup vault svp-prd
key alias/svp-prd-backup
role svp-prd-backup

CloudFront standard logging
svp-cloudfront-prd

EventBridge svp-security-* rules
forwarder role

On OU Prod: SCP **svp-prod-guardrails**, Backup policy **svp-backup-prd**

OU NonProd

svp-stg

VPC flow logs

AMG svp-stg
only during a session
ADMIN svp-platform-admins
VIEWER svp-platform-viewers

WAF web ACL logging
aws-waf-logs-svp

AWS Backup vault svp-stg
key alias/svp-stg-backup
role svp-stg-backup

CloudFront standard logging
svp-cloudfront-stg

EventBridge svp-security-* rules
forwarder role

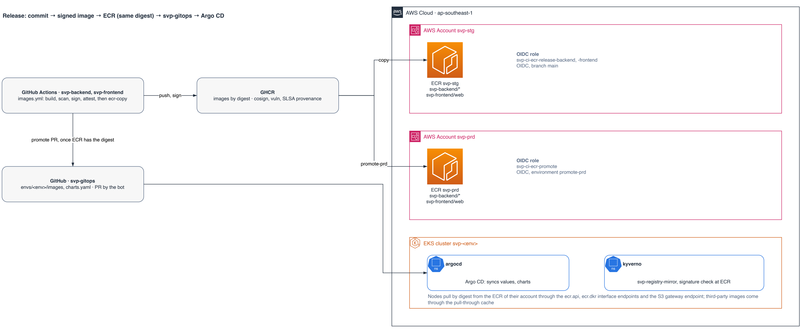
Off in svp-stg (aws/paid-features.json):
- Security Hub
- Config recorder
- Inspector ECR

GuardDuty and CloudTrail run in every account

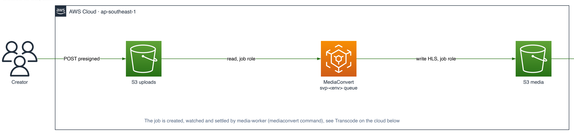
On OU NonProd: Backup policy **svp-backup-stg**

Flows — image release, upload to playback, request to the web and API, telemetry, transcode, backup, restore

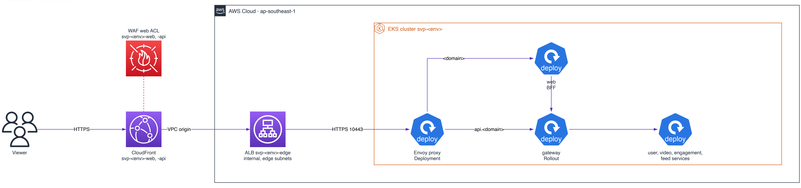
Release: commit → signed image → ECR (same digest) → svp-gitops → Argo CD



Upload to playback

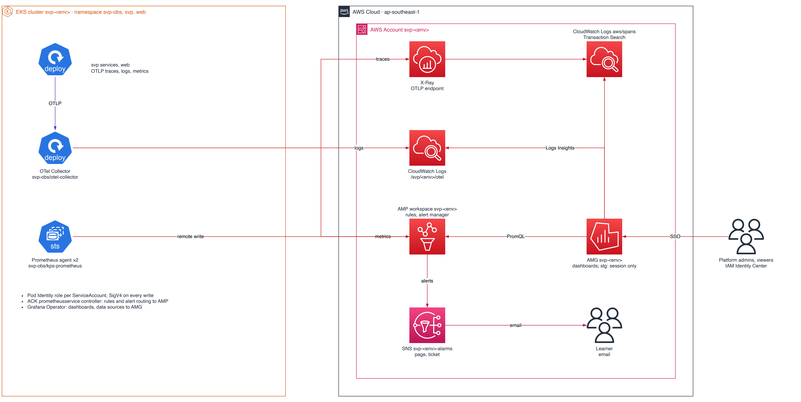


Request to the web and API

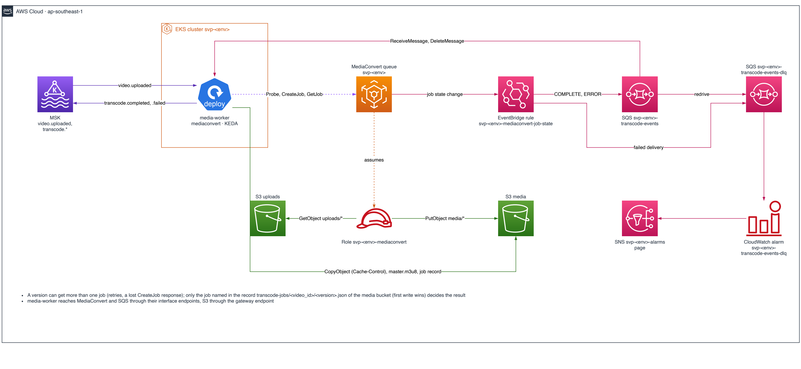


CloudFront adds CloudFront-Viewer-Address and the gateway trusts it only on connections from the VPC; web forwards it to the gateway · the ALB accepts only CloudFront (security group on the origin-facing prefix list) · access logs of the distributions (none for media on the flat-rate Free plan) and the web ACL logs go to the log archive account

Telemetry: Collector and Prometheus agent to AMP, CloudWatch Logs and X-Ray, read in AMG (eks)

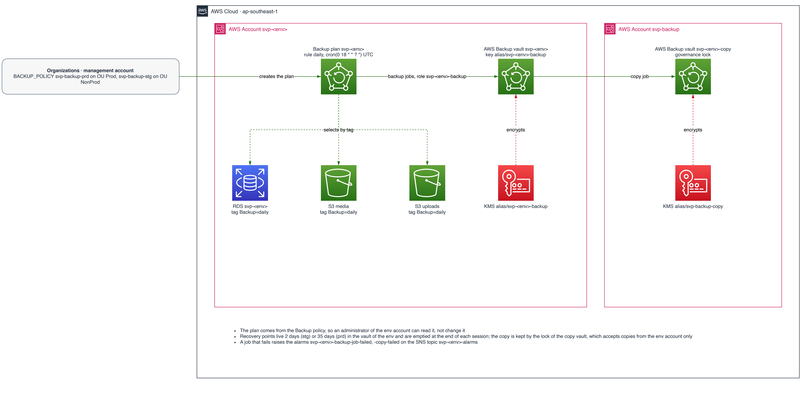


Transcode on the cloud (eks): media-worker, MediaConvert, EventBridge, SQS



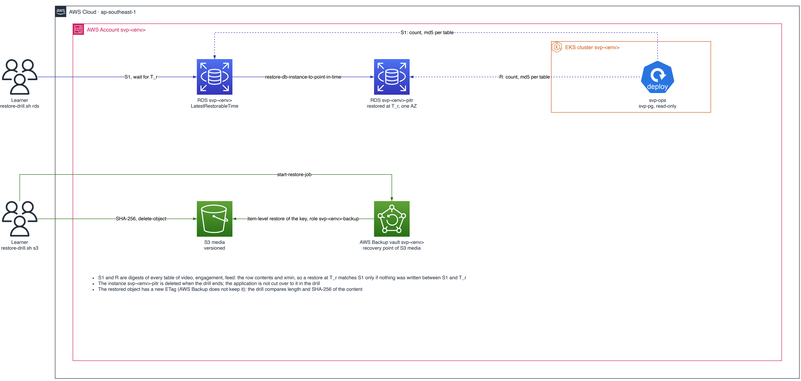
A version can get more than one job (retries, a lost CreateJob response), only the job named in the record transcode-jobs/<video_id>/<version>.json of the media bucket (first write wins) decides the result
media-worker reaches MediaConvert and SQS through their interface endpoints, S3 through the gateway endpoint

Backup: Backup policy of the organization, plan and jobs in the env account, copy to the vault of svp-backup



The plan comes from the Backup policy, so an administrator of the env account can read it, not change it
Recovery points live 2 days (stg) or 35 days (prd) in the vault of the env and are emptied at the end of each session; the copy is kept by the lock of the copy vault, which accepts copies from the env account only
A job that fails raises the alarms svp-<env>-backup-job-failed, -copy-failed on the SNS topic svp-<env>-alarms

Restore drills: PITR of RDS, one object of S3 media from a recovery point



S1 and R are digests of every table of video, engagement, feed: the row contents and xmin, so a restore at T_r matches S1 only if nothing was written between S1 and T_r
The instance svp-<env>-pit· is deleted when the drill ends; the application is not cut over to it in the drill
The restored object has a new ETag (AWS Backup does not keep it): the drill compares length and SHA-256 of the content